

1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

Definition 1.10.1

The **Kronecker product** of an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$ is the $mp \times nq$ block matrix $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$. The **vec operator** places the n columns of an $m \times n$ matrix one below another, giving a vector of length mn : $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$.

Proposition 1.10.1

For matrices of compatible sizes:

- (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ (mixed-product rule);
- (b) $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ for nonsingular $\mathbf{A}, \mathbf{B}$. More generally $\mathbf{A}^- \otimes \mathbf{B}^-$ is a generalized inverse of $\mathbf{A} \otimes \mathbf{B}$;
- (c) $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B})$ and $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$;
- (d) if $\mathbf{A}$ ($m \times m$) and $\mathbf{B}$ ($p \times p$) are symmetric with eigenvalues λ_i and μ_j , the eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are the products $\lambda_i \mu_j$, and $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p (\det \mathbf{B})^m$;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f) $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ and $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$.

Proof.

- (a) The (i, j) block of the left side is $\sum_k a_{ik} c_{kj} \mathbf{B}$, which is the (i, j) block of the right side.
- (b) Transposition is direct, and the rest follows from (a), for example $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \otimes \mathbf{BB}^- \mathbf{B}$.
- (b) With $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^\top$ and $\mathbf{B} = \mathbf{Q}\mathbf{M}\mathbf{Q}^\top$, (a) and (b) give $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{\Lambda} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$, a spectral decomposition because $\mathbf{P} \otimes \mathbf{Q}$ is orthogonal and $\mathbf{\Lambda} \otimes \mathbf{M}$ is diagonal with entries $\lambda_i \mu_j$. The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$ is idempotent with the same rank as $\mathbf{A} \otimes \mathbf{B}$ (Proposition 1.3.3(a)). Its trace is $\text{tr}(\mathbf{AA}^-) \text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$ by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The j th column of $\mathbf{ABC}$ is $\mathbf{ABc}_j = \sum_k c_{kj} \mathbf{Ab}_k$, which is the j th block of $(\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$. The trace identity is $\sum_{ij} a_{ij} b_{ij}$ written two ways.

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The determinant formula in (d) holds for all square $\mathbf{A}$ and $\mathbf{B}$, not only symmetric ones. We need only the symmetric case.

Example. The balanced two-way layout

Let observations y_{ij} , $i = 1, \dots, a$, $j = 1, \dots, b$, be stacked with j running inside i . The additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because $\mathbf{I}_a \otimes \mathbf{1}_b$ has a 1 in column i exactly for the b observations in row i . With $\bar{\mathbf{J}}_k = k^{-1} \mathbf{J}_k$, a symmetric idempotent averaging matrix, the mixed-product rule shows that $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$, $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$ and $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ replace each observation by its row mean, its column mean and the grand mean. The matrix $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ is symmetric and idempotent, with trace $a + b - 1 = \text{rank}(\mathbf{X})$. It is the projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 16.1.2 in Chapter 16). The script checks this for $a = 3$ and $b = 4$, together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F")          # stack the columns

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B))          # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)),      # intercept
               np.kron(np.eye(a), one(b)),   # row indicators
               np.kron(one(a), np.eye(b))])  # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)
```

[Run this code online](#)

1.10.1 Exercises

1.10.1.1 A. Check your understanding

Exercise. A1

With observations stacked as in Example (The balanced two-way layout), the row and column indicator matrices are $\mathbf{R} = \mathbf{I}_a \otimes \mathbf{1}_b$ and $\mathbf{C} = \mathbf{1}_a \otimes \mathbf{I}_b$. Use the mixed-product rule to compute $\mathbf{R}^\top \mathbf{R}$, $\mathbf{C}^\top \mathbf{C}$ and $\mathbf{R}^\top \mathbf{C}$, and identify the cross-product matrix $\mathbf{X}^\top \mathbf{X}$ of the balanced additive model.

Solution. By Proposition 1.10.1(a) and (b), $\mathbf{R}^\top \mathbf{R} = (\mathbf{I}_a^\top \mathbf{I}_a) \otimes (\mathbf{1}_b^\top \mathbf{1}_b) = b \mathbf{I}_a$ and $\mathbf{C}^\top \mathbf{C} = (\mathbf{1}_a^\top \mathbf{1}_a) \otimes \mathbf{I}_b = a \mathbf{I}_b$, while $\mathbf{R}^\top \mathbf{C} = (\mathbf{I}_a^\top \mathbf{1}_a) \otimes (\mathbf{1}_b^\top \mathbf{I}_b) = \mathbf{1}_a \mathbf{1}_b^\top$, the $a \times b$ matrix of ones. With $\mathbf{1}_{ab} = \mathbf{1}_a \otimes \mathbf{1}_b = \mathbf{R} \mathbf{1}_a = \mathbf{C} \mathbf{1}_b$, the cross-product matrix of $\mathbf{X} = [\mathbf{1}_{ab}, \mathbf{R}, \mathbf{C}]$ is

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} ab & b \mathbf{1}_a^\top & a \mathbf{1}_b^\top \\ b \mathbf{1}_a & b \mathbf{I}_a & \mathbf{1}_a \mathbf{1}_b^\top \\ a \mathbf{1}_b & \mathbf{1}_b \mathbf{1}_a^\top & a \mathbf{I}_b \end{pmatrix}.$$

Every cell of the layout is filled once, so each pair of a row level and a column level occurs

exactly once, which is what the block of ones records.

Exercise. A2

Show that $(\mathbf{A} \otimes \mathbf{B})^\top (\mathbf{A} \otimes \mathbf{B}) = (\mathbf{A}^\top \mathbf{A}) \otimes (\mathbf{B}^\top \mathbf{B})$, and deduce that the Kronecker product of two matrices with orthonormal columns again has orthonormal columns. Show also that $\|\mathbf{A}\|_F = \|\text{vec}(\mathbf{A})\|$, so that vec turns the Frobenius norm into the Euclidean norm.

Exercise. A3

Let $\Sigma = \Sigma_1 \otimes \Sigma_2$ with Σ_1 ($m \times m$) and Σ_2 ($p \times p$) positive definite — a *separable* covariance matrix, one variance structure across rows and another across columns. Show that Σ is positive definite, that $\Sigma^{-1} = \Sigma_1^{-1} \otimes \Sigma_2^{-1}$, that $\Sigma^{1/2} = \Sigma_1^{1/2} \otimes \Sigma_2^{1/2}$ in the sense of Theorem 1.7.6, and that $\log \det \Sigma = p \log \det \Sigma_1 + m \log \det \Sigma_2$.

1.10.1.2 B. Practice

Exercise. B1

Let $\mathbf{Y}$ be $n \times q$, let $\mathbf{X}$ be $n \times p$ with full column rank and let $\mathbf{B}$ be $p \times q$, so that q responses are regressed on the same $\mathbf{X}$. Using Proposition 1.10.1(f), show that $\text{vec}(\mathbf{XB}) = (\mathbf{I}_q \otimes \mathbf{X}) \text{vec}(\mathbf{B})$ and that minimizing $\|\mathbf{Y} - \mathbf{XB}\|_F^2$ leads to

$$(\mathbf{I}_q \otimes \mathbf{X}^\top \mathbf{X}) \text{vec}(\mathbf{B}) = \text{vec}(\mathbf{X}^\top \mathbf{Y}).$$

Deduce that a multivariate regression on a common model matrix is q separate univariate regressions, one per column of $\mathbf{Y}$.

Solution. Part (f) with $\mathbf{XB}\mathbf{I}_q$ gives $\text{vec}(\mathbf{XB}) = (\mathbf{I}_q^\top \otimes \mathbf{X}) \text{vec}(\mathbf{B})$. By Exercise (A2), $\|\mathbf{Y} - \mathbf{XB}\|_F^2 = \|\text{vec}(\mathbf{Y}) - (\mathbf{I}_q \otimes \mathbf{X}) \text{vec}(\mathbf{B})\|^2$, an ordinary least squares criterion with model matrix $\mathbf{I}_q \otimes \mathbf{X}$. Its normal equations are

$$(\mathbf{I}_q \otimes \mathbf{X})^\top (\mathbf{I}_q \otimes \mathbf{X}) \text{vec}(\mathbf{B}) = (\mathbf{I}_q \otimes \mathbf{X})^\top \text{vec}(\mathbf{Y}),$$

and the mixed-product rule turns the left side into $(\mathbf{I}_q \otimes \mathbf{X}^\top \mathbf{X}) \text{vec}(\mathbf{B})$ and the right side into $\text{vec}(\mathbf{X}^\top \mathbf{Y})$. The matrix $\mathbf{I}_q \otimes \mathbf{X}^\top \mathbf{X}$ is block diagonal with identical blocks, so the system separates into $\mathbf{X}^\top \mathbf{X} \mathbf{b}_j = \mathbf{X}^\top \mathbf{y}_j$ for each column $\mathbf{y}_j$ of $\mathbf{Y}$. Nothing is gained by treating the responses jointly, as long as they share a model matrix and the criterion is the unweighted sum of squares.

Exercise. B2

Show that $\mathbf{A} \otimes \mathbf{B} = \mathbf{0}$ iff $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$, and that if $\mathbf{A} \otimes \mathbf{B} = \mathbf{C} \otimes \mathbf{D}$ with $\mathbf{A}, \mathbf{C}$ of one size and $\mathbf{B}, \mathbf{D}$ of another, all four nonzero, then $\mathbf{C} = c^{-1}\mathbf{A}$ and $\mathbf{D} = c\mathbf{B}$ for some scalar $c \neq 0$. A separable covariance matrix therefore determines its two factors only up to reciprocal scalars.

Solution. The (i, j) block of $\mathbf{A} \otimes \mathbf{B}$ is $a_{ij}\mathbf{B}$, so the product is zero iff every $a_{ij}\mathbf{B} = \mathbf{0}$. If $\mathbf{A} \otimes \mathbf{B} = \mathbf{C} \otimes \mathbf{D}$, then $a_{ij}\mathbf{B} = c_{ij}\mathbf{D}$ for all i, j . Choose k, l with $a_{kl} \neq 0$. If $c_{kl} = 0$ then $\mathbf{B} = \mathbf{0}$, which is excluded, so $\mathbf{D} = c\mathbf{B}$ with $c = a_{kl}/c_{kl} \neq 0$. Substituting back, $a_{ij}\mathbf{B} = c c_{ij}\mathbf{B}$ for all i, j , and $\mathbf{B} \neq \mathbf{0}$ forces $a_{ij} = c c_{ij}$, that is, $\mathbf{C} = c^{-1}\mathbf{A}$. Conversely $(c^{-1}\mathbf{A}) \otimes (c\mathbf{B}) = \mathbf{A} \otimes \mathbf{B}$.

Exercise. B3

Write $\bar{\mathbf{J}}_k = k^{-1}\mathbf{J}_k$ and $\mathbf{Q}_k = \mathbf{I}_k - \bar{\mathbf{J}}_k$, both symmetric idempotent, of ranks 1 and $k - 1$. Expanding $\mathbf{I}_{ab} = (\bar{\mathbf{J}}_a + \mathbf{Q}_a) \otimes (\bar{\mathbf{J}}_b + \mathbf{Q}_b)$, show that the four matrices

$$\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b, \quad \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b, \quad \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b, \quad \mathbf{Q}_a \otimes \mathbf{Q}_b$$

are symmetric idempotent, sum to $\mathbf{I}_{ab}$ and multiply to zero in pairs, with ranks 1, $a - 1$, $b - 1$ and $(a - 1)(b - 1)$. Show that the matrix $\mathbf{M}$ of [Example \(The balanced two-way layout\)](#) is the sum of the first three, so that $\mathbf{I} - \mathbf{M} = \mathbf{Q}_a \otimes \mathbf{Q}_b$. These are the grand mean, row, column and interaction parts of a two-way analysis of variance, and their ranks are its degrees of freedom.

Solution. Each of the four is a Kronecker product of symmetric idempotent matrices, hence symmetric idempotent by [Proposition 1.10.1\(e\)](#), with rank the product of the ranks by [Proposition 1.10.1\(c\)](#). Expanding the product of the two decompositions of the identity gives $\mathbf{I}_a \otimes \mathbf{I}_b$ as their sum. Any two of them have a factor $\bar{\mathbf{J}}\mathbf{Q} = \mathbf{0}$ or $\mathbf{Q}\bar{\mathbf{J}} = \mathbf{0}$ in one of the two slots, so their product vanishes by the mixed-product rule. The four ranks add to $1 + (a - 1) + (b - 1) + (a - 1)(b - 1) = ab$, as Theorem 1.9.3 requires. Finally

$$\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b = (\bar{\mathbf{J}}_a + \mathbf{Q}_a) \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes (\bar{\mathbf{J}}_b + \mathbf{Q}_b) - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b,$$

which collects into $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b$, of rank $1 + (a - 1) + (b - 1) = a + b - 1$. Subtracting from $\mathbf{I}_{ab}$ leaves $\mathbf{Q}_a \otimes \mathbf{Q}_b$.

1.10.1.3 C. Going deeper

Exercise. C1

Show that for each m, n there is exactly one $mn \times mn$ matrix $\mathbf{K}_{m,n}$ with $\mathbf{K}_{m,n} \text{vec}(\mathbf{A}) = \text{vec}(\mathbf{A}^\top)$ for every $m \times n$ matrix $\mathbf{A}$, that it is a permutation matrix, and that $\mathbf{K}_{m,n}^\top = \mathbf{K}_{m,n}^{-1} = \mathbf{K}_{n,m}$. Then show that for $\mathbf{A}$ of size $m \times n$ and $\mathbf{B}$ of size $p \times q$,

$$\mathbf{K}_{p,m}(\mathbf{A} \otimes \mathbf{B}) = (\mathbf{B} \otimes \mathbf{A})\mathbf{K}_{q,n}.$$

This **commutation matrix** is how the two orders of a Kronecker product are related, and how derivatives with respect to a matrix are rearranged.

Solution. The map $\text{vec}(\mathbf{A}) \mapsto \text{vec}(\mathbf{A}^\top)$ permutes the mn coordinates: the entry a_{ij} , which sits in position $i + (j - 1)m$ of $\text{vec}(\mathbf{A})$, sits in position $j + (i - 1)n$ of $\text{vec}(\mathbf{A}^\top)$. A linear map on $\mathbb{R}^{mn}$ is determined by its values on the coordinate vectors, and every coordinate vector is $\text{vec}(\mathbf{e}_i \mathbf{e}_j^\top)$ for some i, j , so $\mathbf{K}_{m,n}$ exists and is unique, and it is the permutation matrix just described. Applying the rule twice returns $\text{vec}(\mathbf{A})$, so $\mathbf{K}_{n,m}\mathbf{K}_{m,n} = \mathbf{I}$; a permutation matrix

is orthogonal, so its inverse is its transpose.

For the last identity, let $\mathbf{C}$ be $q \times n$, so that both sides may be applied to $\text{vec}(\mathbf{C})$. By [Proposition 1.10.1\(f\)](#), $(\mathbf{A} \otimes \mathbf{B}) \text{vec}(\mathbf{C}) = \text{vec}(\mathbf{BCA}^\top)$, a $p \times m$ matrix, so the left side sends $\text{vec}(\mathbf{C})$ to $\text{vec}((\mathbf{BCA}^\top)^\top) = \text{vec}(\mathbf{AC}^\top \mathbf{B}^\top)$. The right side sends it to $(\mathbf{B} \otimes \mathbf{A}) \text{vec}(\mathbf{C}^\top) = \text{vec}(\mathbf{AC}^\top \mathbf{B}^\top)$ by the same rule. The two matrices agree on every $\text{vec}(\mathbf{C})$, that is, on all of $\mathbb{R}^{nq}$, so they are equal.

Exercise. C2

In a balanced two-way layout with random row and column effects, the response vector has covariance

$$\mathbf{V} = \sigma_\alpha^2(\mathbf{I}_a \otimes \mathbf{J}_b) + \sigma_\beta^2(\mathbf{J}_a \otimes \mathbf{I}_b) + \sigma^2 \mathbf{I}_{ab}, \quad \sigma^2 > 0.$$

Using the four idempotent matrices of [Exercise \(B3\)](#), show that

$$\mathbf{V} = (b\sigma_\alpha^2 + a\sigma_\beta^2 + \sigma^2) \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + (b\sigma_\alpha^2 + \sigma^2) \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b + (a\sigma_\beta^2 + \sigma^2) \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b + \sigma^2 \mathbf{Q}_a \otimes \mathbf{Q}_b,$$

read off the eigenvalues of $\mathbf{V}$ with their multiplicities, and write down $\det \mathbf{V}$ and $\mathbf{V}^{-1}$. On which subspace does $\mathbf{V}$ act as $\sigma^2 \mathbf{I}$?

Solution. Since $\mathbf{J}_k = k\bar{\mathbf{J}}_k$ and $\mathbf{I}_k = \bar{\mathbf{J}}_k + \mathbf{Q}_k$,

$$\mathbf{I}_a \otimes \mathbf{J}_b = b(\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + \mathbf{Q}_a \otimes \bar{\mathbf{J}}_b), \quad \mathbf{J}_a \otimes \mathbf{I}_b = a(\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{Q}_b),$$

and $\mathbf{I}_{ab}$ is the sum of all four. Collecting coefficients gives the displayed form. The four matrices are symmetric idempotent with pairwise products zero and sum $\mathbf{I}_{ab}$, so by Theorem 1.9.2(a) the columns of an orthonormal basis of each column space are eigenvectors of $\mathbf{V}$, with eigenvalue the coefficient in front. The eigenvalues are therefore $b\sigma_\alpha^2 + a\sigma_\beta^2 + \sigma^2$ once, $b\sigma_\alpha^2 + \sigma^2$ with multiplicity $a - 1$, $a\sigma_\beta^2 + \sigma^2$ with multiplicity $b - 1$ and σ^2 with multiplicity $(a - 1)(b - 1)$. The determinant is their product,

$$\det \mathbf{V} = (b\sigma_\alpha^2 + a\sigma_\beta^2 + \sigma^2)(b\sigma_\alpha^2 + \sigma^2)^{a-1}(a\sigma_\beta^2 + \sigma^2)^{b-1}(\sigma^2)^{(a-1)(b-1)},$$

and the inverse replaces each coefficient in the displayed sum by its reciprocal, since the four terms multiply to zero in pairs and each is idempotent. Finally $\mathbf{V}\mathbf{u} = \sigma^2 \mathbf{u}$ for every $\mathbf{u} \in \mathcal{C}(\mathbf{Q}_a \otimes \mathbf{Q}_b)$: on the interaction space the row and column variance components have no effect at all, which is why that space supplies the error term of the balanced two-way analysis of variance.